Model-assisted estimators with wall-to-wall auxilliary information

Estimator	Point estimator	Variance estimator	Comments
One-phase	$\frac{1}{n}\sum_{x\in s} Y(x)$	$\frac{1}{n(n-1)}\sum_{x\in s}\left(Y(x)-\bar{Y}_s\right)^2$	- No auxiliary information - Exactly unbiased
Difference estimator/external model	$\frac{1}{A_T}\int_T \hat{Y}(x)dx + \frac{1}{n}\sum_{x\in s} R(x)$	$\frac{1}{n(n-1)}\sum_{x\in s}\left(R(x)-\bar{R}_s\right)^2$	- Model fit on an *independent* dataset (another survey, past years etc.) - Works with any model type - Exactly unbiased
Plug-in model-assisted estimator	$\frac{1}{A_T}\int_T \hat{Y}(x)dx + \frac{1}{n}\sum_{x\in s} \hat{R}(x)$	$\frac{1}{n(n-1)}\sum_{x\in s}\left(\hat{R}(x)-\bar{\hat{R}}_s\right)^2$	- Pretend model fit is external and use difference estimator - Asymptotically unbiased for many model types
Classical regression estimator / GREG	Let $\hat{Y}(x)=\mathbf{Z}(x)^\top\hat{\boldsymbol{\beta}}_\mathbf{s}$ $\bar{\mathbf{Z}}^\top\hat{\boldsymbol{\beta}}_\mathbf{s}$ $\bar{\mathbf{Z}} := \frac{1}{A_T}\int_T \mathbf{Z}(x)dx$ (i.e. *exhaustive* auxiliary mean)	*plug-in* : (see above) *g-weight* : $\bar{\mathbf{Z}}^\top\hat{\Sigma}_{\hat{\boldsymbol{\beta}}_s}\bar{\mathbf{Z}}$ $\hat{\Sigma}_{\hat{\boldsymbol{\beta}}_s} := \mathbf{A}_s^{-1}\left[\frac{1}{n^2}\sum_{x\in s}\hat{R}(x)^2\mathbf{Z}(x)\mathbf{Z}(x)^t\right]\mathbf{A}_s^{-1}$ $\mathbf{A}_s := \frac{1}{n}\sum_{x\in s}\mathbf{Z}(x)\mathbf{Z}(x)^\top$.	- Linear assisting model - *g-weight* accounts for fitting the model internally (i.e. using the same plots we use for estimation) by Taylor approximation - Mean residual is zero by construction when intercept is included - Asymptotically unbiased - *g-weight* has better properties (e.g. calibration) **Special cases:** • Intercept only = one-phase • Categorical predictors = poststratification • Categorical/continuous interactions = regression within strata
Design-based small area estimator	$\bar{\mathbf{Z}}_G^\top\hat{\boldsymbol{\beta}}_\mathbf{s} + \frac{1}{n_G}\sum_{s_G}\hat{R}(x)$ $\bar{\mathbf{Z}}_G := \frac{1}{A_G}\int_G \mathbf{Z}(x)dx$ (i.e. *exhaustive* aux. mean in G)	*plug-in* : $\frac{1}{n_G(n_G-1)}\sum_{x\in s_G}\left(R(x)-\bar{R}_{s_G}\right)^2$ *g-weight* : $\bar{\mathbf{Z}}_G^\top\hat{\Sigma}_{\hat{\boldsymbol{\beta}}_s}\bar{\mathbf{Z}}_G + \frac{1}{n_G(n_G-1)}\sum_{s_G}\left(\hat{R}(x)-\bar{\hat{R}}_{s_G}\right)^2$	- Fit linear model with large sample s (stabilize cofficient estimates) and apply predictions to small area, denoted G. - "Small area" refers to limited sample size not surface area. - Asymptotically unbiased **Clever Hack :** Using an indicator variable for the small area G as a predictor in the linear model will make the mean residual zero by construction. The point and variance estimator become $\bar{\mathbf{Z}}_G^\top\hat{\boldsymbol{\beta}}_\mathbf{s}$ and $\bar{\mathbf{Z}}_G^\top\hat{\Sigma}_{\hat{\boldsymbol{\beta}}_s}\bar{\mathbf{Z}}_G$ respectively.

IMPORTANT NOTE! These estimators are derived under uniform independent sampling, whereas most NFIs use systematic grids. Based on simulations, using independent-sampling variance estimators under systematic sampling does not bias the point estimator unless there is a periodic structure in the forest interacting with the grid (possibly from regular plantations etc). However, the variance is likely overestimated, thus producing overly wide confidence intervals.